

Neelam Venkata Prasad Akula

CONTACT INFORMATION

Department of Mathematics, FO35
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RESEARCH INTERESTS

My research interests center around Graph Representation Learning (GRL) and Topological Machine Learning/Data Analysis. I am particularly interested in leveraging topological information to build robust models with interpretable learning frameworks.

Recent works have focused on task transferability in GRL, seeking to improve generalization of learned representations across tasks and domains. I hold further interest in applications of GRL within trade and financial network analysis.

EDUCATION

University of Texas at Dallas

Ph.D. Mathematics, Fall 2023 - Present

- Ph.D. Candidate since January 2025
- Advisor: [Dr. Baris Coskunuzer](#)

M.S. Mathematics, Dec 2025

University of Maryland, College Park

B.S. Mathematics and Computer Science, May 2023

- Thesis: [Quiver Representations and Auslander-Reiten Theory](#)
- Advisor: [Dr. Amin Gholampour](#)

PAPERS AND PUBLICATIONS

1. Neelam Akula, Surbhi Kumar, Murat Kantarcioglu, Baris Coskunuzer, (Accepted, 2026) *Same Graph Cross-Task Transfer in GNNs: Protocols and Predictors*. Proceedings of the 43rd International Conference on Machine Learning (**ICML**)
2. Neelam Akula, Varun Shiralkar, Baris Coskunuzer, (Accepted, 2026) *Node2Graph: Diagnosing Task Unification in Graph Learning*. Proceedings of the 30th Pacific-Asia Conference on Knowledge Discovery and Data Mining (**PAKDD**)
3. Neelam Akula, Varun Shiralkar, Murat Kantarcioglu, Baris Coskunuzer, *Graph Label Alignment: A Diagnostic Atlas for Graph Classification*. 2026 (Submitted)
4. Neelam Akula, Rayaans Ansari, Muhammed Ali Yildirim, Baris Coskunuzer, *Input-Output Network Topology Reveals Persistent Structural Regimes in the Global Economy*. (In Preparation)
5. N.V.P. Akula, *Quiver Representations and Auslander-Reiten Theory*. Bachelor's Thesis. 2023
6. N.V.P Akula, *The knitting algorithm for quivers of type AD*. Washington Undergraduate Mathematics Journal. 2023

TALKS

Node2Graph: Diagonalising Task Unification in Graph Learning. Advances in Knowledge Discovery and Data Mining: 30th Pacific-Asia Conference, PAKDD 2026

Graph Label Alignment: A Diagnostic Atlas for Graph Classification, University of Texas at Dallas, April 2, 2025

Task Transferability in Graph Representation Learning, UTD, Richardson, TX, Dec. 2025

Brain Network Modeling, UTD, Richardson, TX, Sep. 2025

Quiver Representations and their Applications, II, UMD, College Park, MD, Dec. 2022

Quiver Representations and their Applications, I, UMD, College Park, MD, Dec. 2022

Bounding Conjugacy Classes for Finite Groups, UMD, College Park, MD, Nov. 2022

On Conjugacy Classes of Varying p -regularity Level, JMU, Harrisonburg, VA, Oct. 2022

Investigating Repeating Decimals, UMD, College Park, MD, Dec. 2021

CONFERENCES AND WORKSHOPS *43rd International Conference on Machine Learning, ICML 2026*, Seoul, South Korea, July 6-11, 2026

Advances in Knowledge Discovery and Data Mining: 30th Pacific-Asia Conference, PAKDD 2026, Hong Kong, China, June 9-12, 2026

Stanford Graph Learning Workshop, Stanford University, October 14, 2025

Shenandoah Undergraduate Mathematics and Statistics Conference, James Madison University, October 29, 2022

Mathematics Opportunities in Research and Education, Virginia Tech, October 7-8, 2022

14th International Conference on Communication Systems & Networks, Bengaluru, India, January 3-8, 2022

TEACHING &
SERVICE

Mentoring:

Varun Shiralkar (UTD)

Rayaan Ansari (UTD)

Teaching:

Linear Algebra (TA, UTD)

Spring 2026

Calculus I (TA, UTD)

Fall 2025

Calculus I (TA, UTD)

Summer 2025

Linear Algebra (TA, UTD)

Spring 2025

Linear Algebra (TA, UTD)

Fall 2024

Calculus II (TA, UTD)

Spring 2024

Linear Algebra (TA, UTD)

Fall 2023

Quiver Representations (Instructor, UMD)

Spring 2023

Algebra I & II (TA, UMD)

Spring 2023

Algebra I & II (TA, UMD)

Fall 2022

Algebra I & II (TA, UMD)

Spring 2022

Algebra I & II (TA, UMD)

Fall 2021

ORGANIZATIONS	Algebra and Combinatorics Seminar (UTD)	Fall 2023 - Present
	Geometry, Topology, Dynamical Systems Seminar (UTD)	Fall 2023 - Present
	Logic Seminar (UMD)	Spring 2023
	Math Club (UMD)	2021 - 2023
PROFESSIONAL MEMBERSHIPS	American Mathematical Society	2024 - Present
	Society for Industrial and Applied Mathematics	2025 - Present
SKILLS	Programming: C/C++, Java, Python, Ocaml, Haskell, MATLAB, R ML Frameworks: PyTorch (Geometric), Keras, Scikit-Learn Algebra Systems: GAP, Magma, SageMath Tools: Git, Unix/Linux, HPC, Cuda, $\text{\TeX}$, $\mathcal{A}\mathcal{M}\mathcal{S}$	
CITIZENSHIP	USA	